Three-step equations

1 Solve the following equations:

a $3(4s+1)=-21$

b $3(2x+1)=15$

c $2(3t-7)=16$

d $-5(3x+8)=-10$

e $2(-3g+7)=-16$

f $-2(3x-5)=28$

g $-3(-2h-5)=33$

h $-3(-2x+7)=-45$

i $4(5x-1)=-24$

j $-4(5x+6)=-104$

k $2(7+2x)=38$

l $-3(4-x)=-12$

m $5(2x-6)=100$

n $6(3x+15)=-72$

o $9(8-x)=45$

p $10(7-3x)=20$

q $20=2(3x+11)$

r $16=4(2x+9)$

s $18=3(2x+14)$

t $-5=5(3x+1)$

u $26=13(8-4x)$

v $32=8(16-5x)$

2 Ryan attempted to solve the equation $9(4x-6)=18$. His working is shown below:

$$\begin{aligned} 9(4x-6) &= 18 \\ 4x-6 &= 9 \\ 4x &= 15 \\ x &= \frac{15}{4} \end{aligned}$$

a What was his mistake?

b Solve the equation correctly.

3 Solve the following equations:

a $3(w+8)+5=44$

b $9(x-7)+8=26$

c $-5(g+4)+9=-41$

d $-7(x+5)+4=-17$

e $6(q+1)-5=-23$

f $4(x+6)-8=48$

g $-8(x+4)-3=-59$

h $-2(x-2)-7=-29$

i $2(x-9)-4=16$

j $5(x-3)-7=-52$

4 Solve the following equations:

a $-\frac{y}{3}+10=17$

b $-\frac{u}{4}+15=8$

c $\frac{5x}{8}-9=-4$

d $\frac{8c}{3}+5=-11$

e $\frac{5x}{\text{[illegible]}}+11=21$

f $\frac{-3c}{\text{[illegible]}}+5=14$

g $-\dfrac{3x}{4}+5=-7$

h $\dfrac{3x}{4}-5=-11$

i $\dfrac{4x}{7}-6=-14$

j $-\dfrac{5x}{6}-6=14$

5 Solve the following equations:

a $\dfrac{x-9}{5}+4=7$

b $\dfrac{x-3}{5}-8=-7$

c $\dfrac{x+2}{3}-25=-22$

d $\dfrac{x+16}{3}+5=3$

e $\dfrac{2x-12}{3}=0$

f $\dfrac{3x+6}{2}=15$

g $\dfrac{8x+4}{5}=-4$

h $\dfrac{-3t-6}{7}=-3$

i $\dfrac{-13-4r}{3}=-15$

j $\dfrac{-9+5x}{2}=-22$

6 Solve the following equations:

a $\dfrac{2\left(x-4\right)}{3}=8$

b $\dfrac{5\left(x+1\right)}{2}=10$

c $\dfrac{-3\left(x-3\right)}{4}=0$

d $\dfrac{6\left(x+5\right)}{7}=12$

e $\dfrac{9\left(x+10\right)}{2}=-6$

f $\dfrac{4\left(x-2\right)}{3}=-3$

g $\dfrac{7\left(x-9\right)}{5}=14$

h $\dfrac{2\left(x+11\right)}{9}=8$

7 Solve the following equations:

a $2\left(\dfrac{x}{3}+1\right)=6$

b $5\left(\dfrac{x}{2}+7\right)=60$

c $4\left(\dfrac{x}{5}-3\right)=-12$

d $3\left(\dfrac{x}{4}-10\right)=15$

e $18=2\left(\dfrac{x}{5}-7\right)$

f $40=10\left(\dfrac{x}{6}-1\right)$

g $8\left(4+\dfrac{x}{2}\right)=48$

h $-4\left(12-\dfrac{x}{3}\right)=-64$